QUANT SERIES · PART 1 OF 4

Stochastic Calculus from Zero

Part 1: probability, properly,
from coin tosses to martingales



Rates with Bip

STOCHASTIC CALCULUS FROM ZERO · PART 1 OF 4

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Part 1: Probability, properly, from coin tosses to martingales, with Python

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Stochastic Calculus from Zero. Part 1: Probability, properly, from coin tosses to martingales, with Python.

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The coin world in this book (a share at 100 that each toss moves up 20% or down 20%, a 60% chance of heads and a deposit growing 2% a step) is made up for teaching, and every random path and toss is simulated with a fixed seed. There is no market data.

The history in Chapters 1, 2, 6 and 7 is checked against the original publications where they are online (Numdam, the Académie des sciences homage to Doeblin, Crossref records) and otherwise against secondary sources; sources are listed at the end of each chapter.

Every number in the text, tables and figures is computed by the companion Python notebooks (numpy only: Part 1 needs no QuantLib), one per chapter, which reproduce each result. Every chapter's exercises are in its notebook too, and the code solutions check themselves.

This book is for education only. It is not financial, investment, legal or tax advice.

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How to use this book

This book follows the eight video episodes of Part 1 of *Stochastic Calculus from Zero*, one chapter per episode, and goes further in each: where the videos show a picture, the book gives the definition and, where it is short, the proof. It assumes calculus at school level and the normal distribution; *Options from Zero* is a good warm-up but not required. By the end of Part 1 you will be able to say precisely what information, conditional expectation, a martingale and a change of measure are, and to check every one of them by hand on a small example and in code.

One puzzle runs through the whole series. Chapter 1 shows the chain rule from school failing on a random path, by exactly the sum of the squared steps. Fixing it is what stochastic calculus is for. Part 1 builds the probability we need, on just eight possible worlds of a coin; Part 2 builds Brownian motion and the Itô integral and fixes the chain rule.

- **Chapter 1** poses the puzzle: the chain rule breaks on a random path.
- **Chapter 2** sets up probability properly: worlds, weights, events, random variables, expectation.
- **Chapter 3** is a maths pit stop on the law of large numbers and the central limit theorem.
- **Chapters 4–6** are the heart of Part 1: information as blocks of worlds (filtrations), conditional expectation, and martingales.
- **Chapter 7** studies the random walk and the Markov property.
- **Chapter 8** changes the odds: the Radon-Nikodym derivative, the discrete seed of Girsanov's theorem.

Each chapter ends with its one idea to remember, which the videos call *Bip's rule*, and a question whose answer opens the next chapter. Two parts of the back matter matter as much as the chapters: **the exercises, with full answers** (five per chapter, three by hand and two in code), and the Part 1 checkpoint.

The coin world

Chapters 2 and 4–8 work on one small example, made up for teaching. A share starts at 100. Each of three tosses of a coin multiplies it by 1.2 (heads) or 0.8 (tails), so there are eight possible worlds, from HHH to TTT. Heads has a 60% chance. A bank deposit grows 2% a step. Eight worlds are few enough to write every one down, and big enough to show every idea of Part 1. Chapters 1, 3 and 7 also simulate random paths and tosses, always with a fixed seed so the numbers repeat.

Numbers you can reproduce

Every number in the text, tables and figures is computed by a companion Python notebook, one per chapter; nothing is typed in by hand. Part 1 needs only numpy, pandas and matplotlib.

The notebooks are free to download from github.com/RatesWithBip/curriculum, in the `stochastic-calculus-from-zero` folder, and each has an *Open in Colab* button. Each notebook ends with its chapter's exercises; the code solutions are collapsed and check themselves with `assert`s.

Reading alongside

This book is written fresh from its own notebooks, but the subject has excellent textbooks, and Part 1 follows the same logical order they do. Good companions are Blitzstein and Hwang's *Introduction to Probability* (free online) for Chapters 2–3 and 5, and Shreve's *Stochastic Calculus for Finance I* for Chapters 4–8. The appendix lists them with the chapters they pair with.

Free, and staying free

The videos, the notebooks and this book are free. If they help you, you can support the channel with a coffee at ko-fi.com/rateswithbip.

This book is for education only and is not financial advice.

CHAPTER 1 · VIDEO EPISODE 1

Why Ordinary Calculus Breaks

Add up the small steps of a smooth path and the chain rule works. Do the same on a random path and it fails, by exactly the sum of the squared steps, which adds up to the time elapsed instead of vanishing.

The rule from school

For a smooth function of time $x(t)$, the chain rule says $d(x^2) = 2x\,dx$. In practice that means: cut the interval $[0, T]$ into n small steps, take x at the start of each step, and add up

$$\sum_{i=0}^{n-1} 2\,x_{t_i}\,(x_{t_{i+1}} - x_{t_i}) \;\approx\; x_T^2 - x_0^2.$$

As the steps shrink, the approximation becomes exact. Try it on the smooth path $x(t) = t + 0.3\sin(2\pi t)$ over one year, which wiggles but has no corners and ends at $x_1 = 1$:

The chain rule on a smooth path: the leftover, the sum of squared steps, vanishes.

STEPS	Σ 2X ΔX	Σ (ΔX)²	X² AT THE END
10	0.7281	0.27188	1.0000
100	0.9722	0.02776	1.0000
1,000	0.9972	0.00278	1.0000
10,000	0.9997	0.00028	1.0000

The two columns always add up to x_T^2 exactly (we will see why in a moment); the second, the **sum of squared steps**, shrinks to zero: 0.00028 with 10,000 steps.

One random path breaks it

Now replace x by a Brownian path W (*Options from Zero*, Part 2, Chapter 1): over one year, 1,000 independent normal steps, each with variance $1/1,000$.

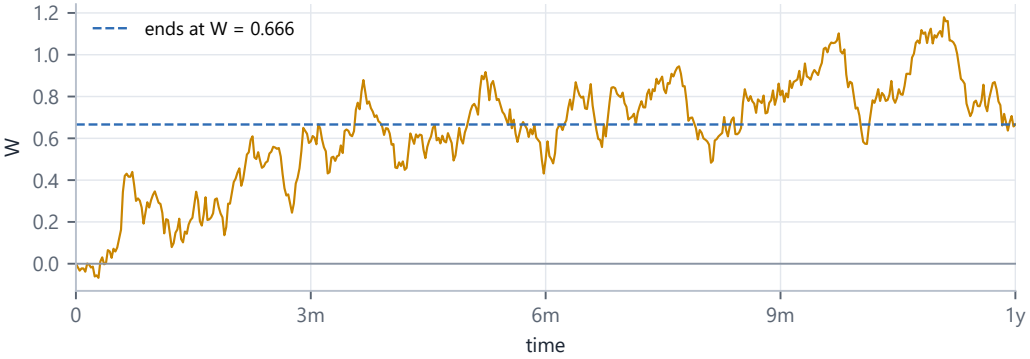


Figure 1.1. One simulated Brownian path, 1,000 steps, seed 2026.

This path ends at $W_1 = 0.666$, so $W_1^2 = 0.444$. The chain-rule sum $\sum 2W\,\Delta W$ comes out at -0.618: not a rounding error, but a miss of 1.062, more than twice the answer.

Where the gap lives

For any two numbers a and b,

$$b^2 - a^2 = 2a\,(b-a) + (b-a)^2.$$

Apply this to every step and add up. On the left the squares telescope, so for **any** path, smooth or not,

$$W_T^2 - W_0^2 = \sum 2W_{t_i}\,\Delta W_i \;+\; \sum (\Delta W_i)^2.$$

This is algebra, not probability: the notebook checks it step by step to within 3.5e-16. The school rule keeps the first sum and drops the second. On our path the dropped sum is 1.062, exactly the gap.

Smooth versus rough

Why does the second sum vanish on one path and not the other? On a smooth path each step is roughly proportional to Δt, so each square is of order $(\Delta t)^2$, and $n = T/\Delta t$ of them add up to something of order Δt, which goes to zero. On a Brownian path each step has standard deviation $\sqrt{\Delta t}$, so each square is of order Δt itself, and n of them add up to something of order T.

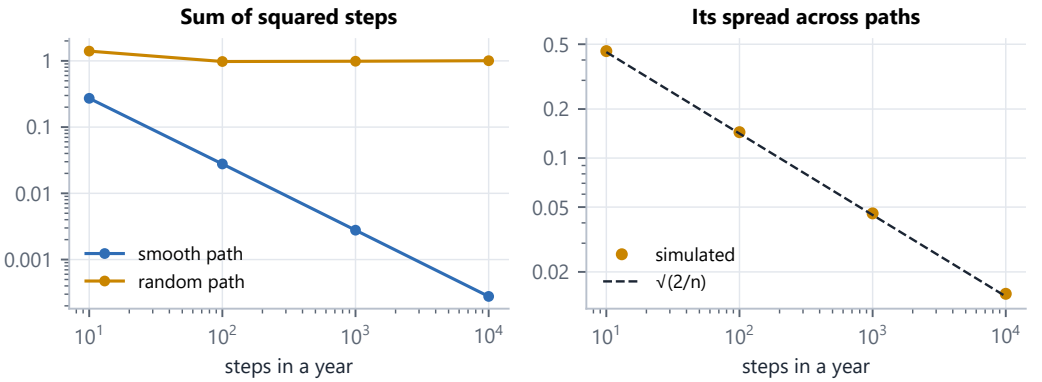


Figure 1.2. Left: the sum of squared steps on the smooth path falls like 1/n; on a 10,000-step Brownian path, measured with coarser and finer steps, it stays near 1. Right: its spread across 1,000 simulated paths, against √(2/n).

Precisely: each $(\Delta W_i)^2$ has mean Δt and variance $2(\Delta t)^2$, and the steps are independent, so

$$E\Big[\sum (\Delta W_i)^2\Big] = T, \qquad \mathrm{Var}\Big[\sum (\Delta W_i)^2\Big] = \frac{2T^2}{n} \to 0.$$

The sum of squared steps does not just *average* T: its spread vanishes, so in the limit it **equals** T on (almost) every path. Part 2 calls this the **quadratic variation** of Brownian motion.

Every path

Over 10,000 paths of 1,000 steps:

W^2 at the end averages its variance, 1; the chain-rule sum averages about 0; the squared steps add up to 1.

AVERAGE OVER ALL PATHS	VALUE
W at the end, squared	0.983
Chain rule: add up 2W·ΔW	-0.018
Sum of squared steps	1.001

The squared steps vary from path to path by only 0.044 (one standard deviation, against the theoretical $\sqrt{2/n}$ = 0.045). The chain-rule sum averages about zero, which is no accident: Part 2 shows it is an Itô integral, and Itô integrals are martingales with mean zero.

A little history

In 1900 Louis Bachelier modelled prices on the Paris Bourse as random in his thesis *Théorie de la spéculation*. In 1923 Norbert Wiener put Brownian motion on a rigorous footing. On 26 February 1940 Wolfgang (Vincent) Doeblin, a soldier in the French army, deposited a sealed envelope with the Académie des sciences, *Sur l'équation de Kolmogoroff*; he died in June 1940, and the envelope stayed shut for 60 years, until 18 May 2000. It contained a change-of-variables formula for diffusions. Meanwhile, in 1944, Kiyosi Itô published *Stochastic integral*. The corrected chain rule of Part 2 is now often called the Itô-Doeblin formula.

The road map

Part 1 builds probability properly on eight coin worlds: information, best guesses, martingales and changes of odds. Part 2 builds Brownian motion and the Itô integral, and in Episode 15 fixes the chain rule by putting the squared steps back. Parts 3 and 4 put it to work: pricing, interest rates and jumps.

Bip's rule. On a random path, the squares don't vanish.

Try this. Run the path for 2 years instead of one. Where does the sum of squared steps settle?

Sources

Companion notebook: `s5e01_why_calculus_breaks.ipynb`. Every number in this chapter is computed by it.

1. Numdam, *Théorie de la spéculation (L. Bachelier), Annales scientifiques de l'École Normale Supérieure, 3e série, tome 17 (1900), 21-86*. https://www.numdam.org/articles/10.24033/asens.476/
2. Université Paris-Saclay (author's page), *J.-M. Bismut, address at the homage to Wolfgang (Vincent) Doeblin: the pli cacheté 'Sur l'équation de Kolmogoroff', deposited with the Académie des sciences on 26 February 1940 (no. 11668), opened 18 May 2000, published in Comptes rendus, série 1, 331 (2000); Doeblin died 21 June 1940*. https://imo.universite-paris-saclay.fr/~jean-michel.bismut/media/files/BBAWAdS%20Bismut.pdf
3. Project Euclid (Crossref record), *Stochastic integral (K. Itô), Proceedings of the Imperial Academy, Tokyo 20(8) (1944), 519-524*. https://doi.org/10.3792/pia/1195572786
4. Wiley (Crossref record), *Differential-Space (N. Wiener), Journal of Mathematics and Physics 2 (1923), 131-174*. https://doi.org/10.1002/sapm192321131

CHAPTER 2 · VIDEO EPISODE 2

Eight Possible Worlds

A probability model is a list of possible worlds with weights. An event is a set of worlds, a random variable is a number attached to each world, and an expectation is a weighted average over the worlds.

Last chapter's question

The answer is 2. Each squared step averages its time step, so over two years the squared steps add up to the time elapsed, and their spread still vanishes as the steps shrink.

The sample space

Toss a coin three times. Each possible outcome, such as HTH, is a **world**, written ω, and the set of all of them is the **sample space** Ω:

$$\Omega = \{\mathrm{HHH}, \mathrm{HHT}, \mathrm{HTH}, \mathrm{HTT}, \mathrm{THH}, \mathrm{THT}, \mathrm{TTH}, \mathrm{TTT}\}.$$

A **probability** P gives each world a weight. Our coin lands heads with chance $p = 0.6$ and the tosses are independent, so a world with k heads has weight $p^k(1-p)^{3-k}$: 0.216 for HHH, 0.144 for HHT, 0.064 for TTT.

Random variables

A **random variable** is a function $X : \Omega \to \mathbb{R}$, one number for each world. Put a share in this world: it starts at $S_0 = 100$ and each toss multiplies it by 1.2 or 0.8. Then S_1, S_2, S_3 , the number of heads, and any payoff are all random variables:

The coin world. The call is struck at 100 on the final price.

WORLD	WEIGHT P	HEADS	S_1	S_2	S_3	CALL PAYS
HHH	0.216	3	120.00	144.00	172.80	72.80
HHT	0.144	2	120.00	144.00	115.20	15.20
HTH	0.144	2	120.00	96.00	115.20	15.20
HTT	0.096	1	120.00	96.00	76.80	0.00
THH	0.144	2	80.00	96.00	115.20	15.20
THT	0.096	1	80.00	96.00	76.80	0.00
TTH	0.096	1	80.00	64.00	76.80	0.00
TTT	0.064	0	80.00	64.00	51.20	0.00

Nothing about a random variable is random. It is a fixed rule; the only random thing is which world we end up in.

Events

An **event** is a set of worlds, and its probability is the total weight of its worlds:

$$P(A) = \sum_{\omega \in A} P(\omega).$$

"The share ends above 100" is the event $\{S_3 > 100\}$, the 4 worlds with at least two heads, with probability 0.648. Its complement, ending below 100, has probability 0.352.

Expectation

The **expectation** of a random variable is its weighted average over all worlds:

$$E_P[X] = \sum_{\omega \in \Omega} P(\omega)\, X(\omega).$$

For the final price, $E_P[S_3] = 112.4864$. A shortcut: each toss multiplies the share by $0.6 \times 1.2 + 0.4 \times 0.8 = 1.04$ on average, and the tosses are independent, so $E_P[S_3] = 100 \times 1.04^3$. No world actually ends at this number: an expectation is an average over the worlds, not a forecast of any one of them. The **variance** $\mathrm{Var}(X) = E[(X - E[X])^2]$ measures the spread; the standard deviation of S_3 is 37.36.

The call's expectation under these weights is 22.2912. Chapter 6 shows why that is *not* its price: pricing needs different weights.

The rules

In 1933, Andrey Kolmogorov's *Grundbegriffe der Wahrscheinlichkeitsrechnung* set out the rules every probability model obeys:

1. $P(A) \geq 0$ for every event A;
2. $P(\Omega) = 1$;
3. if $A_1, A_2, \ldots$ are disjoint, $P(A_1 \cup A_2 \cup \ldots) = P(A_1) + P(A_2) + \ldots$.

With finitely many worlds every set of worlds can be an event and these rules are all there is. With infinitely many worlds, such as all Brownian paths, not every set can be given a probability consistently, and the events have to be a well-behaved collection called a **sigma-algebra**: it contains Ω, and it is closed under complements and countable unions. Chapter 4 meets sigma-algebras again, as information.

Bip's rule. A random variable is just a number for every world.

Try this. What is the chance of exactly two heads?

Sources

Companion notebook: `s5e02_worlds_and_random_variables.ipynb` . Every number in this chapter is computed by it.

1. Springer (Crossref record), *Grundbegriffe der Wahrscheinlichkeitsrechnung (A. Kolmogoroff), Ergebnisse der Mathematik, Springer, Berlin, 1933*. https://doi.org/10.1007/978-3-642-49888-6

CHAPTER 3 · VIDEO EPISODE 3

Maths Pit Stop: Averages Settle Down

The average of many independent tosses settles on the expectation, and the wobble that is left is bell-shaped and shrinks like one over the square root of the number of tosses.

Last chapter's question

Three worlds have exactly two heads, each with weight $0.6^2 \times 0.4$, so the chance is 0.432.

Variance adds up

Score a toss 1 for heads and 0 for tails. One toss has mean p and variance $p(1-p) = 0.24$. For **independent** random variables, variances add:

$$\mathrm{Var}(X_1 + \cdots + X_n) = \mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n),$$

because every cross term $E[(X_i - \mu)(X_j - \mu)]$ with $i \neq j$ is zero. So 100 tosses have variance 24 and standard deviation 4.90 heads, around a mean of 60.

The law of large numbers

The **fraction** of heads in n tosses has mean p and variance $p(1-p)/n$, which goes to zero. By Chebyshev's inequality, $P(|\bar{X}_n - p| > \varepsilon) \leq p(1-p)/(n\varepsilon^2)$, so the fraction of heads settles on p: the (weak) **law of large numbers**.

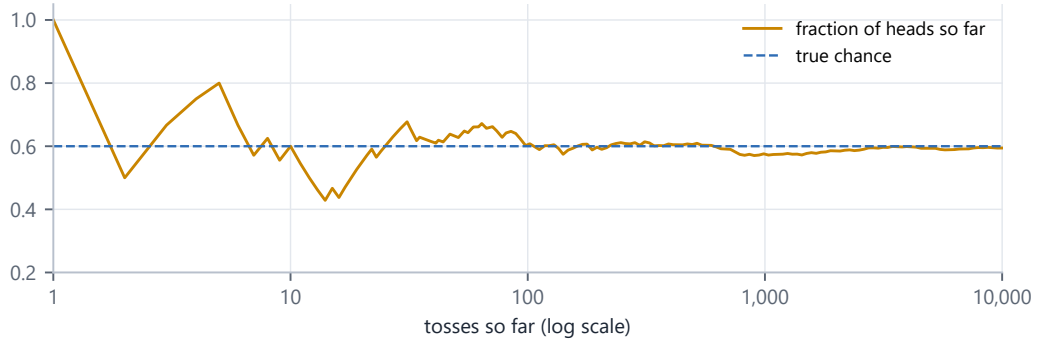


Figure 3.1. One simulated run of 10,000 tosses of the 60% coin: the fraction of heads so far.

In this run the fraction is 0.61 after 100 tosses, 0.574 after 1,000 and 0.5943 after 10,000.

The central limit theorem

The leftover wobble has a shape. Standardise the number of heads, $(\text{heads} - np)/\sqrt{np(1-p)}$, and as n grows its distribution approaches the standard normal: the **central limit theorem**.

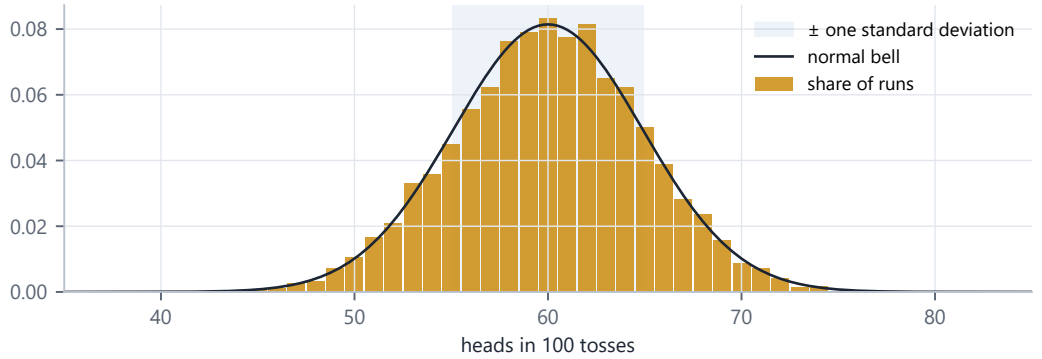


Figure 3.2. 10,000 simulated runs of 100 tosses: the number of heads against the normal bell with the same mean and standard deviation.

Over the 10,000 runs the mean is 60.04 and the standard deviation 4.89. 64.2% of runs land within one standard deviation of 60, a little under the normal curve's 68.3% because heads come in whole numbers (within 4.90 of 60 means 56 to 64), and 95.0% within two.

One over root n

The spread of the fraction of heads is $\sqrt{p(1-p)/n}$:

Four times the tosses, half the spread.

TOSSES	SPREAD, THEORY	SPREAD, 4,000 RUNS
100	0.0490	0.0482
400	0.0245	0.0244
1,600	0.0122	0.0121
6,400	0.0061	0.0063

This is why Monte Carlo pricing needs a hundred times more paths for each extra decimal place. It is also Chapter 1's puzzle seen again: each squared Brownian step is random, but it averages one time step, so by the law of large numbers the total settles on the time elapsed, with a wobble that shrinks like $1/\sqrt{n}$.

The gambler's fallacy

After a run of tails, heads is not "due". The tosses are independent, so the next one is heads with chance 0.6 whatever came before. The average settles because early luck is diluted by later tosses, not because the coin repays it.

Bip's rule. Averages settle; the wobble shrinks like one over root n.

Try this. How many tosses halve the spread of the heads fraction you get from 100?

Companion notebook: `s5e03_averages_settle_down.ipynb`. Every number in this chapter is computed by it.

CHAPTER 4 · VIDEO EPISODE 4

Blocks of Information

What you know at time n is how finely you can tell the worlds apart. Information comes in blocks, the blocks only ever split, and the growing sequence of splits is a filtration.

Last chapter's question

The answer is 400. The spread goes like $1/\sqrt{n}$, so halving it takes four times the tosses.

Blocks of information

Before any toss you cannot tell the eight worlds apart: they sit in one block. After the first toss you know whether the world starts with H or T: two blocks of four. After two tosses, four blocks of two; after three, eight blocks of one.

TIME N	0	1	2	3
Blocks	1	2	4	8

The blocks at time n form a **partition** of Ω: they don't overlap and together they cover every world. Each block at time $n+1$ sits inside a block at time n, so the partitions only get finer.

Filtrations and sigma-algebras

The collection $\mathcal{F}_n$ of all events you can decide at time n, which here means every union of time-n blocks, is a **sigma-algebra** (Chapter 2). The increasing sequence

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \mathcal{F}_3$$

is a **filtration**. It is the mathematical form of "information grows over time and is never forgotten". With finitely many worlds a sigma-algebra and a partition carry the same information, which is why blocks are the right picture.

Known at time n

An event A is **known at time** n, written $A \in \mathcal{F}_n$, when it is a union of whole time-n blocks: once you know your block, you know whether A happened. "The first toss was heads" is known at time 1. "The share ends above 100" cuts through blocks until time 3: after HT, for example, it could still go either way.

The same idea for random variables: X is **$\mathcal{F}_n$-measurable** (known at time n) when it takes a single value on each time-n block. S_2 is $\mathcal{F}_2$-measurable; S_3 is not.

Price versus history

After HT and after TH the share is at the same price, $100 \times 1.2 \times 0.8 = 96$. So S_2 splits the worlds into only 3 groups (144, 96 and 64), while the history splits them into 4. The information generated by the price, written $\sigma(S_2)$, is coarser than $\mathcal{F}_2$:

$$\sigma(S_2) \subsetneq \mathcal{F}_2.$$

Knowing today's price is knowing less than knowing how you got here. With 16 histories but only 5 prices after four tosses, the gap grows quickly.

No peeking

A trading strategy may use only what is known at the time: the number of shares you hold from time n to $n+1$ must be $\mathcal{F}_n$-measurable. Such strategies are called **adapted** (or non-anticipating). The rule looks obvious here; in Part 2 it decides how the Itô integral is built, and why it takes the integrand at the start of each step.

Bip's rule. Information is how finely you can tell the worlds apart.

Try this. With 4 tosses, how many histories are there after the last toss, and how many different prices?

Companion notebook: `s5e04_filtrations.ipynb`. Every number in this chapter is computed by it.

CHAPTER 5 · VIDEO EPISODE 5

The Best Guess

Conditional expectation is the best guess of a random variable using only what you know. It is a random variable itself, one guess per block, and two rules do most of the work: the tower property and taking out what is known.

Last chapter's question

16 histories but only 5 prices: the price depends only on the number of heads.

The definition

With no information, the best single guess of S_3 is its expectation, 112.4864. Once you know which time-n block B you are in, average only over the worlds still possible, with their weights rescaled to add up to one:

$$E[X \mid \mathcal{F}_n](\omega) = \frac{\sum_{\omega' \in B} P(\omega')\, X(\omega')}{\sum_{\omega' \in B} P(\omega')} \qquad \text{for } \omega \in B.$$

This is the **conditional expectation** of X given $\mathcal{F}_n$. It takes one value per block, so it is itself a random variable, and it is $\mathcal{F}_n$-measurable.

Best guesses of the final price after one and two tosses.

WORLD	WEIGHT	S_3	$E[S_3 \mid F_1]$	$E[S_3 \mid F_2]$
HHH	0.216	172.80	129.79	149.76
HHT	0.144	115.20	129.79	149.76
HTH	0.144	115.20	129.79	99.84
HTT	0.096	76.80	129.79	99.84
THH	0.144	115.20	86.53	99.84
THT	0.096	76.80	86.53	99.84
TTH	0.096	76.80	86.53	66.56
TTT	0.064	51.20	86.53	66.56

After heads the best guess is 129.7920, after tails 86.5280. Since each remaining toss multiplies the price by 1.04 on average, $E[S_3 \mid \mathcal{F}_n] = S_n \times 1.04^{3-n}$.

Three rules

For random variables X, Y and times $m \le n$:

1. **Linearity.** $E[aX + bY \mid \mathcal{F}_n] = aE[X \mid \mathcal{F}_n] + bE[Y \mid \mathcal{F}_n]$.
2. **Tower property.** $E\big[E[X \mid \mathcal{F}_n] \mid \mathcal{F}_m\big] = E[X \mid \mathcal{F}_m]$. In particular $E\big[E[X \mid \mathcal{F}_n]\big] = E[X]$.
3. **Taking out what is known.** If Y is $\mathcal{F}_n$-measurable, $E[Y\,X \mid \mathcal{F}_n] = Y\,E[X \mid \mathcal{F}_n]$.

Why the tower property holds. A time-m block is a union of time-n blocks. Averaging X over each small block, then averaging those averages with the small blocks' total weights, is the same as averaging X over the big block directly. On our tree: $0.6 \times 129.79 + 0.4 \times 86.53 = 112.4864 = E[S_3]$.

Why taking out what is known holds. On each block Y is a constant, and constants come out of averages. After heads, $E[S_1 S_3 \mid \mathcal{F}_1] = 15575.04 = 120 \times 129.79$.

Guessing a payoff

The same machine works for any payoff. For the call struck at 100, $E[\text{call} \mid \mathcal{F}_2]$ is 49.76 after HH, 9.12 after HT or TH, and 0 after TT. An option's value over time is a column of this kind, with one change Chapter 6 makes: the weights.

Best in what sense?

Among all $\mathcal{F}_n$-measurable guesses G, the conditional expectation minimises the mean squared error $E[(X - G)^2]$. Write $G = E[X \mid \mathcal{F}_n] + D$ with D $\mathcal{F}_n$-measurable; the cross term $E[(X - E[X \mid \mathcal{F}_n])\,D]$ vanishes by the tower property and taking out what is known, so the error is $E[(X - E[X \mid \mathcal{F}_n])^2] + E[D^2]$, smallest when $D = 0$. Conditional expectation is the **least-squares projection** onto what you know.

A regression on the first toss alone recovers the block averages, up to sampling noise.

BLOCK	BLOCK AVERAGE	LEAST-SQUARES FIT, 100,000 WORLDS
First toss heads	129.79	130.01
First toss tails	86.53	86.67

The typical error of the best guess is 30.8, against 37.4 for the plain average. The same projection idea drives the regression step of American Monte Carlo (*XVA*, Chapter 15).

Bip's rule. Conditional expectation is your best guess with what you know.

Try this. If heads had a 50% chance, what would the best guess of S_3 be after a first head?

Companion notebook: `s5e05_conditional_expectation.ipynb`. Every number in this chapter is computed by it.

CHAPTER 6 · VIDEO EPISODE 6

Martingales: Fair Games

A martingale is a process whose best guess for the next step is where it is now. Under the risk-neutral odds the discounted share is a martingale, and so is the discounted value of any option, whose value today is the cost of copying it.

Last chapter's question

The answer is 120. With a fair coin each toss multiplies the price by $0.5 \times 1.2 + 0.5 \times 0.8 = 1$ on average, so the best guess of S_3 is today's price: a process with no drift.

The definition

A process $M_0, M_1, \ldots$ is a **martingale** with respect to a filtration $(\mathcal{F}_n)$ and a probability P if each M_n is known at time n, has a finite expectation, and

$$E_P[M_{n+1} \mid \mathcal{F}_n] = M_n \qquad \text{for every } n.$$

By the tower property this extends to any horizon: $E[M_m \mid \mathcal{F}_n] = M_n$ for all $m \geq n$, and in particular $E[M_n] = M_0$. A martingale is a fair game: whatever has happened, you expect to break even from here on. If $\leq$ holds instead of $=$ it is a **supermartingale** (an unfavourable game), if $\geq$ a **submartingale**.

The discounted share

A deposit grows 2% a step, so we measure the share in units of the deposit: $\tilde{S}_n = S_n / 1.02^n$. Test the martingale property at all 7 nodes of the tree (1 + 2 + 4; the tree keeps HT and TH apart):

Under the real-world odds the discounted share drifts up at every node; under q it does not drift at all.

NODE	DISCOUNTED NOW	NEXT, REAL ODDS	NEXT, ODDS Q
start	100.0000	101.9608	100.0000
H	117.6471	119.9539	117.6471
T	78.4314	79.9692	78.4314
HH	138.4083	141.1222	138.4083
HT	92.2722	94.0815	92.2722
TH	92.2722	94.0815	92.2722
TT	61.5148	62.7210	61.5148

Under $p = 0.6$ the discounted share grows by $1.04/1.02 = 1.0196$ a step: a submartingale. Under

$$q = \frac{1.02 - 0.8}{1.2 - 0.8} = 0.55$$

it is a martingale at every node. This q is the **risk-neutral probability** of *Options from Zero* (Part 1, Chapter 6), now with a precise meaning: the unique odds that make the discounted share a martingale. They exist and are unique here because $0.8 < 1.02 < 1.2$; if the deposit beat the share in both states, there would be an arbitrage instead.

The option's value

Define the call's value at each node as the q-weighted best guess of its payoff, discounted back to that time:

$$V_n = \frac{E_Q[\text{call} \mid \mathcal{F}_n]}{1.02^{\,3-n}}.$$

Then $V_n/1.02^n = E_Q[\text{call}/1.02^3 \mid \mathcal{F}_n]$ is a Q-martingale by the tower property: any process of the form $E[X \mid \mathcal{F}_n]$ is a martingale.

The call's value process. Today it is worth the price.

WORLD	PAYS AT 3	VALUE AT 2	VALUE AT 1
HHH	72.80	45.9608	28.3987
HHT	15.20	45.9608	28.3987
HTH	15.20	8.1961	28.3987
HTT	0.00	8.1961	28.3987
THH	15.20	8.1961	4.4195
THT	0.00	8.1961	4.4195
TTH	0.00	0.0000	4.4195
TTT	0.00	0.0000	4.4195

Today the call is worth 17.2628; after one toss 28.3987 or 4.4195.

Why that is the price

Hold Δ_0 shares and borrow the rest, with

$$\Delta_0 = \frac{V_1(H) - V_1(T)}{S_1(H) - S_1(T)}.$$

Bought for the option's price, the copy is worth the option's value after either first toss.

THE COPY	VALUE
Shares held	0.5995
Borrowed	42.6853
Worth after heads	28.3987
Worth after tails	4.4195

Repeat at every node and the portfolio pays the call's payoff in every world. Its cost is 17.2628, so any other price for the call is an arbitrage. Discounting the real-world average payoff instead would give 21.0055: Chapter 8 explains exactly how the two are related.

A constant average is not enough

Let $X_1 = +1$ on a first head and -1 on a first tail, and $X_2 = -X_1$. With a fair coin both average 0 at every time, but $E[X_2 \mid \mathcal{F}_1] = -X_1 \neq X_1$: after a head the best guess of X_2 is -1. A martingale must hold its best guess at **every** node, not just on average.

A little history

"Martingale" first named a family of betting systems, such as doubling the stake after every loss. Jean Ville brought the word into probability theory in his 1939 thesis *Étude critique de la notion de collectif*, and Joseph Doob developed much of the theory used today.

Bip's rule. A martingale's best guess for tomorrow is today.

Try this. Under q, is the undiscounted share a martingale?

Sources

Companion notebook: `s5e06_martingales.ipynb`. Every number in this chapter is computed by it.

1. Numdam, *Étude critique de la notion de collectif (J. Ville), thèse, Gauthier-Villars, Paris, 1939*. https://numdam.org/item/THESE_1939__218__1_0/
2. Wikipedia, *Martingale (probability theory), History section: originally a class of betting strategies; the term introduced by Ville (1939); much of the original theory developed by Joseph Leo Doob*. https://en.wikipedia.org/wiki/Martingale_(probability_theory)

CHAPTER 7 · VIDEO EPISODE 7

Random Walks: Only Now Matters

The random walk has independent steps whose variances add and whose squares add up to exactly the number of steps. A Markov process's future depends on its past only through its present, but a payoff can still depend on the path.

Last chapter's question

No. Under q the share grows like the deposit, by a factor of 1.02 a step on average; only the discounted share is a martingale.

The symmetric random walk

Let $X_1, X_2, \ldots$ be independent, each $+1$ or -1 with equal chances, and $M_n = X_1 + \cdots + X_n$ with $M_0 = 0$. This **symmetric random walk** is the simplest random process there is; Part 2 obtains Brownian motion as its limit.

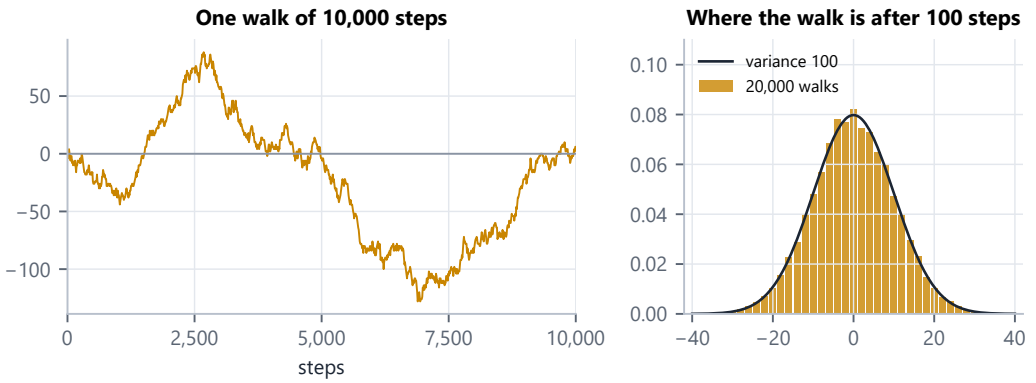


Figure 7.1. Left: one simulated walk of 10,000 steps. Right: where 20,000 walks are after 100 steps, against the normal bell with variance 100.

The walk on the left ends at 6 after wandering from -129 to 90.

Independent increments

For $k > l$, the increment $M_k - M_l = X_{l+1} + \cdots + X_k$ uses only steps after time l, so it is independent of everything up to time l, with mean 0 and variance $k - l$. In simulation: $\mathrm{Var}(M_{100}) = 99.48$ (theory 100) and the last 50 steps alone give 50.22 (theory 50). Brownian motion inherits exactly this property.

Quadratic variation, exactly

Each $X_i^2 = 1$, so the sum of squared steps up to time n is

$$[M]_n = \sum_{i=1}^{n} X_i^2 = n$$

on every path, with no noise: 10,000 after 10,000 steps. Chapter 1's "squared steps add up to the time elapsed" holds here without any averaging.

Two martingales

M_n is a martingale: $E[M_{n+1} \mid \mathcal{F}_n] = M_n + E[X_{n+1}] = M_n$. Its square is not:

$$E[M_{n+1}^2 \mid \mathcal{F}_n] = M_n^2 + 2M_n\,E[X_{n+1}] + E[X_{n+1}^2] = M_n^2 + 1,$$

so M_n^2 drifts up by one each step, and $M_n^2 - n$ is a martingale. In simulation its average stays within noise of zero:

20,000 simulated walks.

AFTER N STEPS	AVERAGE OF M² − N	STANDARD ERROR
10	-0.012	0.095
50	-0.383	0.481
100	-0.520	0.991

Remember the "$-n$". It is the discrete form of the extra term the chain rule was missing in Chapter 1, and in Part 2 it becomes $W_t^2 - t$.

The Markov property

A process is **Markov** if, for any function f,

$$E[f(X_{n+1}) \mid \mathcal{F}_n] = E[f(X_{n+1}) \mid X_n]\,:$$

the future depends on the past only through the present. The coin-world share is Markov: after HT and after TH it is at 96 either way, and the best guess of S_3 is 99.84 in both cases.

Same price, same guess of the final price, but different guesses of the path's highest price.

WORLD	S_2	GUESS OF S_3	HIGHEST SO FAR	GUESS OF THE HIGHEST
HHH	144.00	149.76	144.00	161.28
HHT	144.00	149.76	144.00	161.28
HTH	96.00	99.84	120.00	120.00
HTT	96.00	99.84	120.00	120.00
THH	96.00	99.84	100.00	109.12
THT	96.00	99.84	100.00	109.12
TTH	64.00	66.56	100.00	100.00
TTT	64.00	66.56	100.00	100.00

When the path matters

A lookback option pays on the highest price along the path. After HT the path has already been to 120; after TH only to 100. Same price, different best guesses: 120.00 against 109.12. The price alone is not enough to value this payoff, but the pair (price, running maximum) is Markov. Path-dependent products are handled by tracking the extra state.

Andrey Markov published his first paper on chains of dependent variables, now called Markov chains, in 1906.

Bip's rule. For a Markov process, only where you are matters.

Try this. A fair walk is at +3 after 50 steps. What is the best guess of where it will be after 100?

Sources

Companion notebook: `s5e07_random_walk_markov.ipynb`. Every number in this chapter is computed by it.

1. Wikipedia, *Markov chain, History section: Markov's first paper on the topic, 1906*. https://en.wikipedia.org/wiki/Markov_chain

CHAPTER 8 · VIDEO EPISODE 8

Changing the Odds

The same worlds can carry two sets of weights. Their ratio, the Radon-Nikodym derivative Z, averages to one, turns real-world averages into risk-neutral ones, and over time is a martingale: the discrete seed of Girsanov's theorem.

Last chapter's question

The answer is +3. The remaining steps average zero, so the walk's best guess for any later time is where it is now: it is a martingale.

Two sets of odds

The coin world carries the real-world weights P (heads 0.60) and the risk-neutral weights Q (heads 0.55, Chapter 6). Same worlds, same prices, same payoffs; only the weights differ. HHH drops from 0.216 to 0.166, while TTT rises from 0.064 to 0.091.

The Radon-Nikodym derivative

Define, world by world,

$$Z(\omega) = \frac{Q(\omega)}{P(\omega)}.$$

Z is the **Radon-Nikodym derivative** of Q with respect to P, written dQ/dP. Toss by toss it multiplies by $q/p = 0.9167$ on a head and $(1-q)/(1-p) = 1.1250$ on a tail:

Z is above 1 exactly in the four worlds where the share falls and the call pays nothing.

WORLD	P	Q	Z = Q/P	CALL PAYS
HHH	0.2160	0.1664	0.7703	72.80
HHT	0.1440	0.1361	0.9453	15.20
HTH	0.1440	0.1361	0.9453	15.20
HTT	0.0960	0.1114	1.1602	0.00
THH	0.1440	0.1361	0.9453	15.20
THT	0.0960	0.1114	1.1602	0.00
TTH	0.0960	0.1114	1.1602	0.00
TTT	0.0640	0.0911	1.4238	0.00

Z runs from 0.7703 (HHH) to 1.4238 (TTT). Risk-neutral odds lean towards the bad worlds; the size of that lean is the price of risk.

Two identities

$$E_P[Z] = \sum_\omega P(\omega)\,\frac{Q(\omega)}{P(\omega)} = \sum_\omega Q(\omega) = 1, \qquad E_Q[X] = \sum_\omega Q(\omega)X(\omega) = E_P[ZX].$$

Reweighting moves probability between worlds but never creates any. For the first price, $E_P[ZS_1] = E_Q[S_1] = 102$, the share grown at the deposit rate.

Pricing with the real-world odds

The call's real-world average payoff, discounted, is 21.0055: wrong, because it ignores the price of risk. Weight each world by Z first and you get 17.2628, the price from Chapter 6:

$$\text{price} = \frac{E_Q[\text{payoff}]}{1.02^3} = \frac{E_P[Z \cdot \text{payoff}]}{1.02^3}.$$

Z over time

Reveal Z as the tosses come in: $Z_n = E_P[Z \mid \mathcal{F}_n]$. It starts at 1, moves to 0.9167 or 1.1250 after one toss, then to 0.8403, 1.0312 or 1.2656, and ends at Z. By the tower property, (Z_n) is a P-martingale. Here it is simply a product of per-toss factors:

$$Z_n = \prod_{i=1}^{n} \left(\frac{q}{p}\right)^{\mathbf{1}\{\text{toss } i=H\}} \left(\frac{1-q}{1-p}\right)^{\mathbf{1}\{\text{toss } i=T\}}.$$

In continuous time the same object is $Z_t = \exp(-\theta W_t - \frac{1}{2}\theta^2 t)$, and the statement that reweighting by it turns a Brownian motion with drift into one without is **Girsanov's theorem** (Part 3).

Equivalent measures

Changing the measure never changes what can happen. Here every world has positive weight under both P and Q; in general we need $Q(A) = 0$ exactly when $P(A) = 0$. Such measures are **equivalent**, and only then does Z exist and is strictly positive.

Part 1 in one paragraph

A probability model is a set of worlds with weights. Information is a filtration of blocks of worlds that only ever split. The best guess with given information is the conditional expectation, which obeys the tower property. A martingale is a process whose best guess for tomorrow is today, and under the risk-neutral odds discounted prices are martingales. A change of odds is a reweighting by Z, itself a martingale. Part 2 does all of it on continuous paths.

Bip's rule. Same worlds, different weights.

Try this. If $Z = 1$ in every world, what does that say about p and q?

Companion notebook: `s5e08_changing_the_odds.ipynb` . Every number in this chapter is computed by it.

Exercises with answers

Five exercises per chapter: three by hand and two in code. They are the same exercises as at the end of each chapter's notebook, where the code solutions run and check themselves; try each before reading its answer. "Section" in an exercise means a section of that chapter's notebook.

Chapter 1: Why Ordinary Calculus Breaks

Exercise 1.1 *(by hand).* A path takes two steps: $W_0 = 0$, $W_1 = a$, $W_2 = b$. Check the identity $W_2^2 = \sum 2W\Delta W + \sum(\Delta W)^2$ by expanding both sides.

Answer. Right side: $2 \cdot 0 \cdot a + 2a(b-a) + a^2 + (b-a)^2 = 2ab - 2a^2 + a^2 + b^2 - 2ab + a^2 = b^2$. ✓

Exercise 1.2 *(by hand).* For the smooth path $x(t) = t^2$ on $[0, 1]$ with n equal steps, compute $\sum(\Delta x)^2$. What happens as n grows?

Answer. $\Delta x_i = \frac{(i+1)^2 - i^2}{n^2} = \frac{2i+1}{n^2}$, so $\sum_{i=0}^{n-1}(\Delta x_i)^2 = \frac{1}{n^4}\sum(2i+1)^2 = \frac{1}{n^4} \cdot \frac{n(2n-1)(2n+1)}{3} = \frac{4n^2-1}{3n^3} \approx \frac{4}{3n} \to 0$.

Solution 1.2, from the notebook

```
for n in (10, 100, 1000):
    dx = np.diff(np.linspace(0, 1, n + 1) ** 2)
    assert abs((dx ** 2).sum() - (4 * n ** 2 - 1) / (3 * n ** 3)) < 1e-12
print("sum of squared steps of t^2 = (4n^2 - 1) / (3n^3): checked for n = 10, 100, 1000")
```

What it prints. `sum of squared steps of t^2 = (4n^2 - 1) / (3n^3): checked for n = 10, 100, 1000`

Exercise 1.3 *(by hand).* A Brownian path on $[0, T]$ is cut into n steps. What is the average of $\sum(\Delta W)^2$? Its standard deviation? (Hint: for a normal Z with variance v, $E[Z^2] = v$ and $\mathrm{Var}(Z^2) = 2v^2$.)

Answer. Each step has variance T/n, so the average is $n \cdot T/n = T$ for every n. The steps are independent, so the variances add: $n \cdot 2(T/n)^2 = 2T^2/n$, and the standard deviation is $T\sqrt{2/n}$, which goes to 0.

Exercise 1.4 *(in code)*. Plot the path-to-path spread of $\sum(\Delta W)^2$ against n on log-log axes (use Section 6's `spread`). Fit a straight line: what is its slope, and why?

Solution 1.4, from the notebook

```
n = np.array([r["steps"] for r in spread]); sd = np.array([r["sd"] for r in spread])
slope = np.polyfit(np.log(n), np.log(sd), 1)[0]
plt.figure(figsize=(5, 3)); plt.loglog(n, sd, "o-"); plt.xlabel("steps");
plt.ylabel("spread"); plt.show()
assert -0.55 < slope < -0.45     # spread = sqrt(2/n): slope -1/2
print(f"slope {slope:.3f}: the spread falls like 1/sqrt(n)")
```

What it prints. `slope -0.497: the spread falls like 1/sqrt(n)`

Exercise 1.5 *(in code)*. Redo the chain-rule sum with W taken at the **end** of each step instead of the start: $\sum 2W_{\text{end}}\Delta W$. What does it average over many paths, and how far is it from W_T^2? (This is why Part 2's integral insists on the start of each step.)

Solution 1.5, from the notebook

```
_, Wx, dWx = brownian_paths(5_000, 1_000, 1.0, np.random.default_rng(seed + 1))
right = 2 * (Wx[:, 1:] * dWx).sum(axis=1)
# W_end = W_start + dW, so the right-point sum = left-point sum + 2 x squares = W_T^2 +
squares
assert np.abs(right - (Wx[:, -1] ** 2 + sum_of_squares(dWx))).max() < 1e-9
assert abs(right.mean() - 2) < 0.1
print(f"right-point sum averages {right.mean():.3f} (= 2T): it overshoots W_T^2 by the sum of
squared steps")
```

What it prints. `right-point sum averages 2.000 (= 2T): it overshoots W_T^2 by the sum of squared steps`

Chapter 2: Eight Possible Worlds

Exercise 2.1 *(by hand)*. What is the chance of at least one tail in three tosses?

Answer. Everything except HHH: $1 - 0.6^3 = 1 - 0.216 = 0.784$.

Exercise 2.2 *(by hand)*. What is the expected number of heads, $E_P[\text{heads}]$?

Answer. Each toss adds 0.6 heads on average, so $3 \times 0.6 = 1.8$. (Or add up weight × heads over the 8 worlds.)

Exercise 2.3 *(by hand)*. Is "$S_2 = 96$" an event? Which worlds make it up, and what is its probability?

Answer. Yes: every set of worlds is an event here. $S_2 = 96$ needs one head and one tail in the first two tosses: HTH, HTT, THH, THT, with total weight $2 \times 0.6 \times 0.4 = 0.48$.

Exercise 2.4 *(in code)*. Build the same table for **four** tosses. Check that the weights add up to 1 and that $E_P[S_4] = 100 \times 1.04^4$.

Solution 2.4, from the notebook

```
cw4 = CoinWorld(steps=4)
t4 = cw4.table()
assert len(t4) == 16 and abs(sum(r["P"] for r in t4) - 1) < 1e-12
assert abs(cw4.expect(lambda w: cw4.price(w, 4)) - 100 * 1.04 ** 4) < 1e-9
print(len(t4), "worlds; E_P[S4] =", round(cw4.expect(lambda w: cw4.price(w, 4)), 4))
```

What it prints. `16 worlds; E_P[S4] = 116.9859`

Exercise 2.5 *(in code)*. Compute $E_P[\text{call}]$ and the probability that the call pays anything.

Solution 2.5, from the notebook

```
e_call = cw.expect(lambda w: max(cw.price(w, 3) - K, 0))
p_pays = cw.expect(lambda w: float(cw.price(w, 3) > K))
assert abs(p_pays - 0.648) < 1e-12
print(f"E_P[call] = {e_call:.4f}; chance it pays = {p_pays:.3f}")
```

What it prints. `E_P[call] = 22.2912; chance it pays = 0.648`

Chapter 3: Maths Pit Stop: Averages Settle Down

Exercise 3.1 *(by hand)*. In the coin world, S_1 is 120 (chance 0.6) or 80 (chance 0.4). Find its mean, variance and standard deviation.

Answer. Mean $0.6 \times 120 + 0.4 \times 80 = 104$. Variance $0.6 \times 16^2 + 0.4 \times 24^2 = 153.6 + 230.4 = 384$. Standard deviation $\sqrt{384} \approx 19.6$.

Exercise 3.2 *(by hand)*. What is the standard deviation of the number of heads in 400 tosses of the 60% coin?

Answer. Variance $400 \times 0.24 = 96$, so the standard deviation is $\sqrt{96} \approx 9.80$.

Exercise 3.3 *(by hand)*. After five tails in a row, what is the chance the next toss is heads?

Answer. Still 0.6: the tosses are independent. The law of large numbers works by dilution (a run of tails becomes a small part of a long history), not by the coin making up for it.

Exercise 3.4 *(in code)*. Using Section 6's table, fit a line to log(spread) against log(tosses). What slope do you get?

Solution 3.4, from the notebook

```
x = np.log([r["tosses"] for r in wobble]); y = np.log([r["sd_sim"] for r in wobble])
slope = np.polyfit(x, y, 1)[0]
assert -0.55 < slope < -0.45
print(f"slope {slope:.3f}: the wobble shrinks like 1/sqrt(n)")
```

What it prints. `slope -0.491: the wobble shrinks like 1/sqrt(n)`

Exercise 3.5 *(in code)*. Average S_1 over 50 simulated tosses, and repeat 5,000 times. Is the spread of those averages close to $19.6/\sqrt{50}$?

Solution 3.5, from the notebook

```
s1 = np.where(rng.random((5_000, 50)) < p, 120.0, 80.0).mean(axis=1)
assert abs(s1.mean() - 104) < 0.2 and abs(s1.std() / (math.sqrt(384) / math.sqrt(50)) - 1) <
0.05
print(f"mean {s1.mean():.2f}, spread {s1.std():.3f} vs theory {math.sqrt(384 / 50):.3f}")
```

What it prints. `mean 104.05, spread 2.795 vs theory 2.771`

Chapter 4: Blocks of Information

Exercise 4.1 *(by hand)*. List the blocks of information made by "number of heads so far" at time 2. How many are there, and how do they compare with $\mathcal{F}_2$?

Answer. Three groups: {HH·} (2 heads), {HT·, TH·} (1 head), {TT·} (0 heads). It is coarser than $\mathcal{F}_2$, which has four blocks: like the price, the count of heads forgets the order.

Exercise 4.2 *(by hand)*. Is the event "$S_1 > S_2$" known at time 2? At time 1?

Answer. $S_1 > S_2$ means the second toss was tails: worlds HT· and TT·. Those are whole blocks at time 2, so it is known at time 2. At time 1 the block H·· contains both HH· and HT·, so it is not known yet.

Exercise 4.3 *(by hand)*. With four tosses, how many blocks are there at times 0, 1, 2, 3, 4?

Answer. 1, 2, 4, 8, 16: each toss splits every block in two.

Exercise 4.4 *(in code)*. Use `known_at` to find, for every event "the share ends at price x", the first time it is known.

Solution 4.4, from the notebook

```
first = {}
for x in sorted({round(cw.price(w, 3), 6) for w in cw.worlds}, reverse=True):
    ev = {w for w in cw.worlds if abs(cw.price(w, 3) - x) < 1e-9}
    first[x] = min(n for n in range(4) if known_at(ev, n))
assert first[172.8] == 3 and first[51.2] == 3
print(first)
```

What it prints. `{172.8: 3, 115.2: 3, 76.8: 3, 51.2: 3}`

Exercise 4.5 *(in code)*. How many blocks does the final price S_3 alone make? Check it is coarser than $\mathcal{F}_3$.

Solution 4.5, from the notebook

```
groups = {}
for w in cw.worlds:
    groups.setdefault(round(cw.price(w, 3), 9), []).append(w)
assert len(groups) == 4 < len(cw.blocks(3))
print({k: v for k, v in groups.items()})
```

What it prints. `{172.8: ['HHH'], 115.2: ['HHT', 'HTH', 'THH'], 76.8: ['HTT', 'THT', 'TTH'], 51.2: ['TTT']}`

Chapter 5: The Best Guess

Exercise 5.1 *(by hand)*. Find $E_P[\text{heads} \mid \mathcal{F}_1]$ in both blocks.

Answer. After H: 1 head so far plus $2 \times 0.6 = 1.2$ expected, so 2.2. After T: $0 + 1.2 = 1.2$.

Exercise 5.2 *(by hand)*. Show that $E_P[S_2 \mid \mathcal{F}_1] = 1.04\, S_1$ in both blocks.

Answer. After H: $0.6 \times 144 + 0.4 \times 96 = 86.4 + 38.4 = 124.8 = 1.04 \times 120$. After T: $0.6 \times 96 + 0.4 \times 64 = 57.6 + 25.6 = 83.2 = 1.04 \times 80$.

Exercise 5.3 *(by hand)*. Check the tower property at time 2 for the block H: average $E[S_3 \mid \mathcal{F}_2]$ over HH and HT.

Answer. $0.6 \times 149.76 + 0.4 \times 99.84 = 89.856 + 39.936 = 129.792 = E[S_3 \mid \mathcal{F}_1]$ on H. ✓

Exercise 5.4 *(in code)*. Compute $E_P[\text{call} \mid \mathcal{F}_2]$ for the call struck at 100.

Solution 5.4, from the notebook

```
call = lambda w: max(cw.price(w, 3) - 100, 0)
gc = cw.given(2, call)
assert abs(gc["HH"] - (0.6 * 72.8 + 0.4 * 15.2)) < 1e-9 and gc["TT"] == 0
print({b: round(v, 4) for b, v in gc.items()})
```

What it prints. `{'HH': 49.76, 'HT': 9.12, 'TH': 9.12, 'TT': 0.0}`

Exercise 5.5 *(in code).* Repeat Section 7's least-squares fit with indicators for the four time-2 blocks. Do the fitted values match $E[S_3 \mid \mathcal{F}_2]$?

Solution 5.5, from the notebook

```
key = heads[:, 0] * 2 + heads[:, 1]            # 3 = HH, 2 = HT, 1 = TH, 0 = TT
X2 = np.column_stack([key == k for k in (3, 2, 1, 0)]).astype(float)
b2 = np.linalg.lstsq(X2, s3, rcond=None)[0]
for b, v in zip(("HH", "HT", "TH", "TT"), b2):
    assert abs(v - g2[b]) < 0.6
print({k: round(float(v), 2) for k, v in zip(("HH", "HT", "TH", "TT"), b2)})
```

What it prints. `{'HH': 150.04, 'HT': 99.91, 'TH': 99.9, 'TT': 66.45}`

Chapter 6: Martingales: Fair Games

Exercise 6.1 *(by hand).* Check the Q-martingale property of the discounted share at node HT.

Answer. At HT, $S_2 = 96$, discounted $96/1.02^2$. Next step: $0.55 \times 115.2 + 0.45 \times 76.8 = 63.36 + 34.56 = 97.92 = 1.02 \times 96$, discounted by 1.02^3: $96/1.02^2$. ✓

Exercise 6.2 *(by hand).* Is the number of heads minus $0.6n$ a P-martingale?

Answer. Yes: each toss adds 1 with chance 0.6 and 0 otherwise, so the expected increase is 0.6, exactly what $0.6(n+1) - 0.6n$ subtracts.

Exercise 6.3 *(by hand).* In the counterexample, compute $E[X_2 \mid \mathcal{F}_1]$ in both blocks.

Answer. After H, $X_2 = -1$ for sure; after T, $X_2 = +1$. So $E[X_2 \mid \mathcal{F}_1] = -X_1 \neq X_1$.

Exercise 6.4 *(in code).* Check that the discounted call value $V_n/1.02^n$ is a Q-martingale at every node, and **not** a P-martingale.

Solution 6.4, from the notebook

```
for n in range(cw.steps):
    for b in V[n]:
        q_next = (cw.q * V[n + 1][b + "H"] + (1 - cw.q) * V[n + 1][b + "T"]) / R ** (n + 1)
        p_next = (cw.p * V[n + 1][b + "H"] + (1 - cw.p) * V[n + 1][b + "T"]) / R ** (n + 1)
        assert abs(q_next - V[n][b] / R ** n) < 1e-9
        assert p_next > V[n][b] / R ** n or V[n][b] == 0
print("discounted call: Q-martingale at every node; drifts up under P")
```

What it prints. `discounted call: Q-martingale at every node; drifts up under P`

Exercise 6.5 *(in code)*. If the deposit grew 3% a step instead of 2%, what q makes the discounted share a martingale?

Solution 6.5, from the notebook

```
cw3 = CoinWorld(rate=0.03)
assert abs(cw3.q - 0.575) < 1e-12
assert abs(cw3.q * 1.2 + (1 - cw3.q) * 0.8 - 1.03) < 1e-12
print(f"q = {cw3.q:.3f}")
```

What it prints. `q = 0.575`

Chapter 7: Random Walks: Only Now Matters

Exercise 7.1 *(by hand)*. What is $\mathrm{Var}(M_7 - M_3)$ for the fair ±1 walk?

Answer. Four independent steps, each with variance 1: $7 - 3 = 4$.

Exercise 7.2 *(by hand)*. Show that $E[M_{n+1}^2 - (n+1) \mid \mathcal{F}_n] = M_n^2 - n$.

Answer. $M_{n+1}^2 = M_n^2 + 2M_nX_{n+1} + X_{n+1}^2$. Given $\mathcal{F}_n$: M_n is known, $E[X_{n+1}] = 0$, $X_{n+1}^2 = 1$. So the guess is $M_n^2 + 1$, and subtracting $n+1$ leaves $M_n^2 - n$.

Exercise 7.3 *(by hand)*. Is the running maximum of the share, on its own, a Markov process?

Answer. No. Whether the maximum can rise next step depends on how far today's price is below it, which the maximum alone doesn't tell you. The pair (price, running maximum) is Markov.

Exercise 7.4 *(in code)*. Simulate a walk with steps ±2. What is the variance after 100 steps, and the sum of squared steps?

Solution 7.4, from the notebook

```
W2 = np.cumsum(rng.choice([-2, 2], (20_000, 100)), axis=1)
assert abs(W2[:, -1].var() / 400 - 1) < 0.03
print("variance after 100 steps:", round(W2[:, -1].var(), 1), "(theory 400); sum of squared
steps: exactly", 100 * 4)
```

What it prints. `variance after 100 steps: 395.6 (theory 400); sum of squared steps: exactly 400`

Exercise 7.5 *(in code)*. Price a lookback call on the coin world that pays (highest price along the path − 100), using the risk-neutral odds q = 0.55 and 2% interest a step.

Solution 7.5, from the notebook

```
look = lambda w: max(cw.path(w)) - 100
price = cw.expect(look, "Q") / 1.02 ** 3
assert price > cw.expect(lambda w: max(cw.price(w, 3) - 100, 0), "Q") / 1.02 ** 3   # worth
more than the plain call
print(f"lookback price {price:.4f}")
```

What it prints. `lookback price 23.6718`

Chapter 8: Changing the Odds

Exercise 8.1 *(by hand)*. Compute Z for the world HTH.

Answer. $Z = 0.917 \times 1.125 \times 0.917 = (0.55/0.6)^2(0.45/0.4) = 0.9453$.

Exercise 8.2 *(by hand)*. Check that $E_P[Z \cdot S_1] = E_Q[S_1] = 102$.

Answer. Z_1 is 0.917 on H and 1.125 on T: $0.6 \times 0.917 \times 120 + 0.4 \times 1.125 \times 80 = 66 + 36 = 102 = 1.02 \times 100$. ✓

Exercise 8.3 *(by hand)*. What is Z when $p = q$? What would it mean for the share's expected growth?

Answer. $Z = 1$ in every world. Then the real-world odds already make the discounted share a martingale: the share is expected to grow at the deposit rate, so investors are paid nothing extra for its risk.

Exercise 8.4 *(in code)*. Build Z for a 10-toss world. What are its smallest and largest values, and its standard deviation under P?

Solution 8.4, from the notebook

```
cw10 = CoinWorld(steps=10)
z10 = np.array([cw10.weight(w, "Q") / cw10.weight(w, "P") for w in cw10.worlds])
p10 = np.array([cw10.weight(w, "P") for w in cw10.worlds])
assert abs(p10 @ z10 - 1) < 1e-12
sd = math.sqrt(p10 @ (z10 - 1) ** 2)
print(f"Z from {z10.min():.3f} to {z10.max():.3f}; sd under P {sd:.3f}")
```

What it prints. `Z from 0.419 to 3.247; sd under P 0.330`

Exercise 8.5 *(in code).* Draw a random payoff column X (one number per world) and check $E_Q[X] = E_P[ZX]$.

Solution 8.5, from the notebook

```
X = dict(zip(cw.worlds, np.random.default_rng(seed).normal(size=8)))
assert abs(cw.expect(lambda w: X[w], "Q") - cw.expect(lambda w: Z[w] * X[w])) < 1e-12
print("E_Q[X] = E_P[Z X] for a random payoff: checked")
```

What it prints. `E_Q[X] = E_P[Z X] for a random payoff: checked`

Checkpoint, glossary and further reading

Chapter 8's question

If $Z = 1$ in every world then $Q = P$: the real-world odds already make the discounted share a martingale, which happens when $p = q = 0.55$. The share is then expected to grow at exactly the deposit rate, and investors are paid nothing for its risk.

Part 1 checkpoint

Try these before Part 2. The answers follow.

1. In the coin world, compute $E_P[S_2 \mid \mathcal{F}_1]$ in both blocks, then check the tower property.
2. Is $S_n/1.04^n$ a martingale under P? Under Q?
3. For the world THT, give P, Q and Z, and check that $Q = Z \cdot P$.
4. A process has $E[X_n] = 0$ for every n. Does that make it a martingale?

Answers.

1. After heads, $0.6 \times 144 + 0.4 \times 96 = 124.80$; after tails, $0.6 \times 96 + 0.4 \times 64 = 83.20$. Averaging with the time-1 weights gives $0.6 \times 124.80 + 0.4 \times 83.20 = 108.16 = 100 \times 1.04^2 = E_P[S_2]$.
2. Under P, yes: each toss multiplies the share by 1.04 on average, so dividing by 1.04^n removes the drift. Under Q, no: each toss multiplies it by 1.02 on average, so $S_n/1.04^n$ drifts down (a supermartingale).
3. $P = 0.4 \times 0.6 \times 0.4 = 0.096$, $Q = 0.45 \times 0.55 \times 0.45 = 0.111375$, and $Z = Q/P = 1.1602$, which is also $1.125 \times 0.9167 \times 1.125$.
4. No. Chapter 6's $X_1 = \pm 1$, $X_2 = -X_1$ has mean zero at every time, but $E[X_2 \mid \mathcal{F}_1] = -X_1$.

Glossary

Adapted. A process whose value at time n is known at time n. Trading strategies must be adapted: no peeking.

Central limit theorem. A sum of many independent pieces, standardised, is approximately normal.

Conditional expectation. $E[X \mid \mathcal{F}_n]$: the average of X within each block of information; the least-squares best guess of X using what is known at time n.

Equivalent measures. Two probabilities that agree on which events are impossible.

Event. A set of worlds. Its probability is their total weight.

Expectation. The weighted average of a random variable over all worlds.

Filtration. A sequence of sigma-algebras $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \ldots$: information that grows and is never forgotten.

Independent increments. The change of a process over a period is independent of its past. True of the random walk and of Brownian motion.

Known at time n (measurable). An event that is a union of time-n blocks; a random variable constant on each block.

Law of large numbers. The average of many independent copies of a random variable settles on its expectation.

Markov property. The future depends on the past only through the present.

Martingale. A process with $E[M_{n+1} \mid \mathcal{F}_n] = M_n$: the best guess for tomorrow is today. A **submartingale** drifts up ($\geq$), a **supermartingale** down ($\leq$).

Quadratic variation. The sum of squared steps of a path. For the random walk it is exactly n; for Brownian motion it is the elapsed time t.

Radon-Nikodym derivative. $Z = dQ/dP$, the ratio of two equivalent probabilities, world by world. $E_P[Z] = 1$ and $E_Q[X] = E_P[ZX]$.

Random variable. A number attached to each world: a function on the sample space.

Random walk. The sum of independent ± 1 steps.

Risk-neutral probability. The odds Q under which discounted prices are martingales; a pricing tool, not a forecast.

Sample space. The set Ω of all possible worlds.

Sigma-algebra. A collection of events closed under complements and countable unions; with finitely many worlds, the unions of the blocks of a partition.

Tower property. $E\big[E[X \mid \mathcal{F}_n] \mid \mathcal{F}_m\big] = E[X \mid \mathcal{F}_m]$ for $m \le n$.

Variance. $E[(X - E[X])^2]$; for independent random variables, variances add.

Further reading

This book is written from its own notebooks, but these are excellent companions. None of their examples or exercises are used here.

- Joseph K. Blitzstein and Jessica Hwang, *Introduction to Probability* (free online): Chapters 1–4 for Chapter 2, Chapters 4, 5 and 10 for Chapter 3, Chapter 9 for Chapter 5.
- Steven E. Shreve, *Stochastic Calculus for Finance I: The Binomial Asset Pricing Model*: Chapter 2 for Chapters 4–6, Chapter 3 for Chapter 8, Chapter 5 for Chapter 7.
- Salih N. Neftci, *An Introduction to the Mathematics of Financial Derivatives*: a gentle parallel read for the whole of Part 1.
- Steven E. Shreve, *Stochastic Calculus for Finance II: Continuous-Time Models*: the main companion for Parts 2–4.

Check the chapter pointers against the edition you have.